

Rajasthan Board Class 12

Subject – Physics

Time: $3\frac{1}{4}$ Hours

M.M: 56

Question 1:

Write the definition of electric dipole moment.

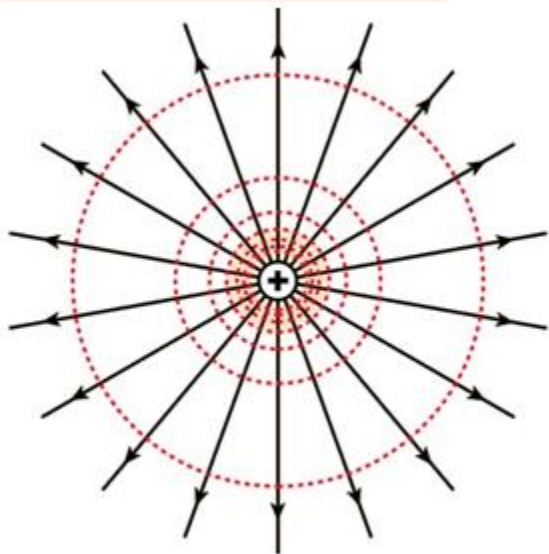
Sol:

Dipole moment vector P of an electric dipole is defined by $p = q \times 2a\hat{p}$. It means it is a vector whose magnitude is charge q times the separation $2a$ (between the pair of charges $q, -q$) and the direction is along the line from $-q$ to q .

Question 2:

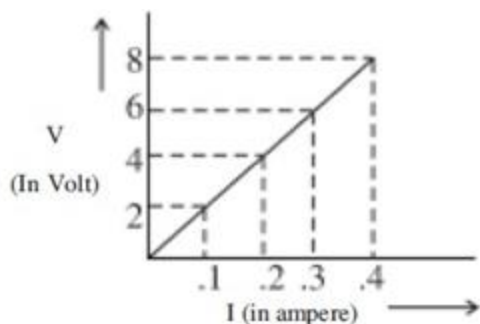
Draw a diagram of equipotential surface for a single charge.

Sol: The diagram of equipotential surface is given below:



Question 3:

Determine the resistance of a resistor with the help of given V-I graph



Sol: The resistance is same as the slope V-I graph. The resistance is

$$\frac{4 - 2}{0.2 - 0.1} = 20\Omega$$

Question 4:

A charge q enters perpendicularly with the direction of a magnetic field B with a velocity V . What would be the force acting on this charge?

Sol:

We have,

$$F = qvB \sin \theta \text{ Where, } \theta \text{ is the angle between of a magnetic field } \vec{B} \text{ with velocity } \vec{v}$$

Since, the charge q enters perpendicularly with the direction of a magnetic field.

So,

$$\theta = 90^\circ$$

Hence, The force acting on the charge q is qvB .

Question 5:

Write the Faraday's law of electromagnetic induction.

Sol:

The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit is known as the Faraday's law of electromagnetic induction.

$$\varepsilon = \frac{-d\phi_B}{dt}$$

Question 6:

A light bulb is rated at 100W for a 220V supply. Find the peak voltage of the source.

Sol:

The peak voltage of the source is,

$$\begin{aligned} V_m &= \sqrt{2V} \\ &= \sqrt{2}(220) \\ &= 311V \end{aligned}$$

Question 7:

Which electromagnetic waves are used in remote controller (switches)?

Sol:

Infrared waves are used in remote controller remote controller.

Question 8:

The radius of curvature of concave mirror is 40 cm. Determine its focal length.

Sol:

We know that the focal length $\frac{R}{2}$,

So, the focal length is

$$\frac{40}{2} = 20cm$$

Question 9:

What is farsightedness or hypermetropia?

Sol:

Hypermetropia (farsightedness) is a common vision condition in which you can see distant objects clearly, but objects nearby may be blurry. The degree of your hypermetropia influence your focusing ability.

Question 10:

How much energy is required to excite a hydrogen atom from its ground state to second excited state?

Sol:

For exciting a hydrogen energy is required as $E_n = \frac{13.6}{n^2} eV$.

And for the second excited state, $n = 2$

The energy is required to excite a hydrogen atom from the ground state is

$$= 13.6 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right]$$

$$= 13.6 \left(1 - \frac{1}{9} \right)$$

$$= 12.08 eV$$

Question 11:

Draw the logic symbol of AND gate

Sol:



Question 12:

Write the two examples of acceptor impurities

Sol:

The examples of acceptor impurities are boron, gallium, and indium. An acceptor impurity accepts an electron from an adjacent atom to create a hole. They are also known as p-type impurities.

Question 13:

Write the SI unit of current density.

Sol:

SI unit of the current density is amp / m^2 .

Question 14:

Write two characteristics of a material to constructs a permanent magnet. Give two examples of such materials

Sol:

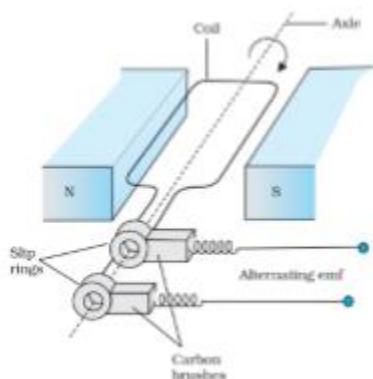
The two characteristics of a material to constructs a permanent magnet are as following:

1. The material should have high retentivity so that the magnet is strong and high coercivity and the magnetization is not erased by stray magnetic fields.
2. The material should have a high permeability

Question 15:

Draw a labeled diagram of an alternating current Generator. Determine the induced electromotive force by the rotation of coil in it.

Sol:



If the coil is rotated with a constant angular speed ω , and θ be the angle between the magnetic field vector B and the area vector A of the coil. At any instant t $\theta = \omega t$. Such that the effective area of the coil exposed magnetic field.

$$\phi = BA \cos \theta = BA \cos \omega t$$

A/C to Faraday's law

$$\begin{aligned}\varepsilon &= -N \frac{d\phi_B}{dt} \\ &= -NBA \frac{d}{dt} (\cos \omega t)\end{aligned}$$

Therefore, the instantaneous value of emf is $\varepsilon = NBA$

Question 16:

Current in a circuit falls from 5.0A to zero in 0.1 S. If an average emf of 100 Volt is induced then calculates self-inductance of a inductor in the circuit.

Sol:

The induced emf of a inductor in the circuit is

$$\begin{aligned}\varepsilon &= -L \frac{dI}{dt} \\ \Rightarrow -100 &= L \frac{5}{0.1} \\ \Rightarrow L &= 2H\end{aligned}$$

Question 17:

What is displacement current? Obtain an expression of displacement current for a charged capacitor. Write Ampere-Maxwell's law.

Sol:

A time varying electric field in a region produces a current is known as called displacement current.

So, charge on parallel plate capacitor $q(t) = \sigma(t) A$

But,

$$\begin{aligned}\sigma(t) &= \varepsilon_0 E(t) \\ \therefore q(t) &= \varepsilon_0 A E(t)\end{aligned}$$

Where,

$\sigma(t)$ is the instantaneous charge per unit area

$E(t)$ is electric flux $\phi E(t)$

$$\therefore q(t) = \epsilon_0 \phi E(t)$$

$$\frac{dq(t)}{dt} = \epsilon_0 \frac{d\phi_E(t)}{dt}$$

Therefore,

Rate of change = $\epsilon_0 \times$ rate of change of electron flux $|\phi E|$

The quantity $\epsilon_0 \frac{d\phi_E(t)}{dt}$ is known as displacement current.

Hence, Ampere's Maxwell law is given $\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i + \frac{d\phi_0}{dt} \right)$

Question 18:

A reflected light becomes completely polarised when corresponding incident angle is 60° . When light is incident on the substance. Determine the velocity of refracted ray in the medium.

Sol:

We have,

$$\mu = \tan i_p$$

$$\Rightarrow \frac{c}{v} = \tan i_p \quad \left[\because \mu = \frac{c}{v} \right]$$

$$\therefore i_p = 60^\circ \text{ and } v = 3 \times 10^8 \text{ m/s}$$

$$\Rightarrow v = \frac{c}{\tan i_p}$$

$$= \frac{3 \times 10^8}{\tan 60^\circ}$$

$$= \sqrt{3} \times 10^8$$

Question 19:

The threshold frequency for cesium metal is 5.16×10^{14} Hz. Determine its work function in eV.

Sol:

Certain minimum amount of energy is required to be given to an electron to pull it out from the surface of the metal. This minimum energy required of an electron is known as work function.

Therefore,

Work function

$$\begin{aligned} &= \frac{6.626 \times 10^{-34} \times 5.16 \times 10^{14}}{1.602 \times 10^{-19}} \\ &= 2.13 \text{ eV} \end{aligned}$$

Question 20:

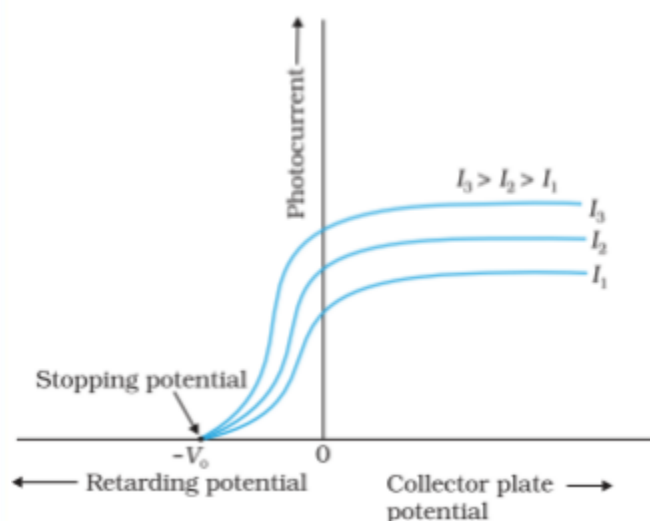
What is stopping voltage (or cut voltage)? Plot a graph of variation of photoelectric current with collector plate potential for two incident radiations of same frequency and different intensities.

Sol:

The minimum negative (retarding) potential V_0 given to the plate A for which the photocurrent stops or becomes zero is called the cut-off or stopping potential.

And,

Graph of variation of photoelectric current with collector plate potential for two incident radiations.



Question 21:

Write the statement of Bohr's Second postulate of quantisation. Determine the wavelength of first spectral line in the Lyman series of the hydrogen

spectrum.[Rydberg constant $R = 1.097 \times 10^7 \text{ m}^{-1}$].

Sol:

A/c to second postulate electron, revolve only the orbit angular momentum

Its integral multiple of $\frac{h}{2\pi}$

Therefore,

For first line Lyman series,

$$\frac{1}{\lambda} = R \left[\frac{1}{1^2} - \frac{1}{2^2} \right]$$

$$\Rightarrow \frac{1}{\lambda} = 1.097 \times 10^7 \left[1 - \frac{1}{4} \right]$$

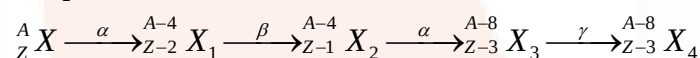
$$\Rightarrow \lambda = 1.21 \times 10^{-7} \text{ m}$$

OR

Write the law of radioactive decay. A radioactive nucleus is decaying in the following way. Determine the mass number and atomic number of final product X_4 When initial nucleus has mass number $A = 238$ and atomic number $Z = 92$.

Sol:

We know that according to the law of radioactive decay rate of sample of large number of nuclei is directly proportional to the number of nuclei present in the sample.



Such that the mass number and the atomic number of the final product are

$$X_4 \text{ is } 238-8=230$$

$$92-3=89$$

Question 22:

Write two drawbacks of Rutherford's atomic model.

Sol:

Drawbacks of Rutherford atomic model are as following:

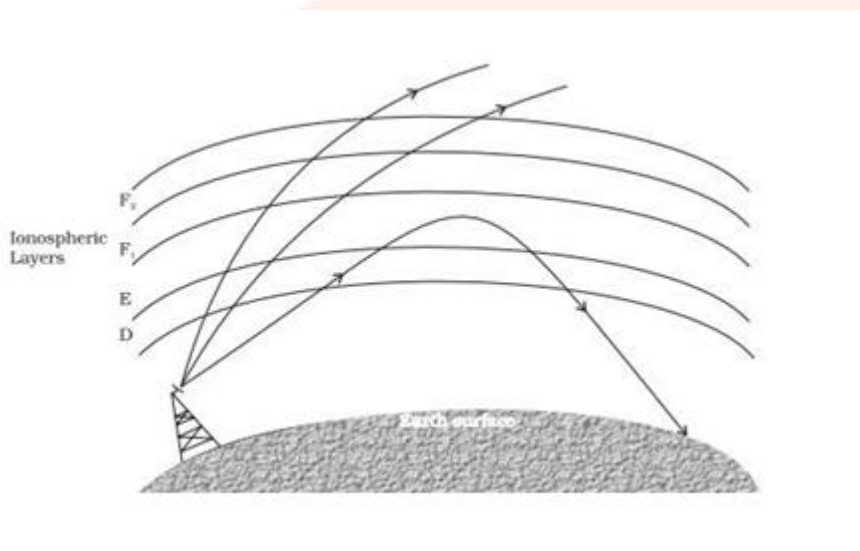
- (1) The model did not explain the stability of atom
- (2) The model did not explain the concept of electronic configuration and energy levels of shell

Question 23:

Explain sky wave propagation by making a diagram.

Sol:

The frequency range from a few MHz up to 30 to 40 MHz, long distance communication can be achieved by ionospheric reflection of radio waves back towards the earth is known as the sky wave propagation and it is used for short wave broadcast



Question 24:

Write definitions

(i) Transducer

Sol:

Device that converts one form of energy into another can be termed as a transducer. And the electrical transducer converts some physical variable (pressure, force, temperature) into corresponding variation in the electrical signal at its output

(ii) Modulation

Sol:

Modulation states that, original low frequency message/information signal cannot be transmitted to long distances. Such that, at the transmitter, information contained in the low frequency message signal is superimposed on a high frequency wave, which acts as a carrier of the information.

Its types are as follows:

- (i) Amplitude Modulation(AM)
- (ii) Frequency Modulation(FM)
- (iii) Phase Modulation(PM)

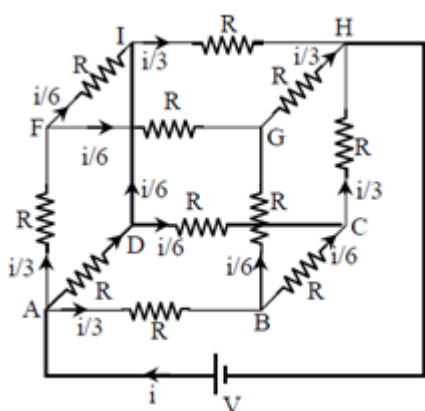
Question 25:

Write Kirchhoff's first rule. A battery of 10V and negligible internal resistance is connected to the diagonally opposite corners of a cubical network consisting of 12 resistors each of resistance 20Ω . Determine the equivalent resistance of the network.

Sol:

The Kirchhoff's first law states that the algebraic sum of the currents into any junction is zero.

$$\sum i = 0$$



Now,

Apply KVL in loop ABCHA then we get,

$$-\frac{i}{3}R - \frac{i}{6}R - \frac{i}{3}R + V = 0$$

$$\Rightarrow V = \frac{5}{6}iR$$

$$\Rightarrow \frac{V}{i} = \frac{5}{6}R$$

$$\Rightarrow R_{eq} = \frac{5}{6}(20) \quad [\because R = 20\Omega]$$

$$= \frac{5}{3}\Omega$$

Question 26:

Why reflecting telescope is superior in comparison to refracting telescope? Write two

reasons.

Magnifying power of a telescope is 8. When it is adjusted for parallel rays the distance between eyepiece and objective lens is 18cm. determine the focal length of both the lenses.

Sol:

Here is no chromatic aberration in reflecting telescope

And the spherical aberration can remove in reflecting telescope.

Such that the magnifying power of a telescope for parallel rays adjustment $\frac{f_0}{f_e}$

$$\therefore \frac{f_0}{f_e} = 8$$

$$\Rightarrow f_0 = 8f_e$$

Now,

The distance between eyepiece and the objective lens is 18cm

$$f_0 + f_e = 18$$

$$\therefore 8f_e + f_e = 18 \quad [\because f_0 = 8f_e]$$

$$\Rightarrow 9f_e = 18$$

$$\Rightarrow f_e = 2\text{cm}$$

Hence,

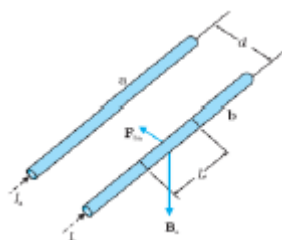
$$f_e = 2\text{cm in } f_0 + f_e = 18\text{cm}$$

$$\therefore f_0 = 16\text{cm}$$

Question 27:

Determine the force acting between two parallel current carrying conductor wires.
Write theoretical definition of ampere on this basis.

Sol:



We have seen in the above figure that two long parallel conductors a and b separated by a distance d and carrying (parallel) currents I_a and I_b , respectively. The conductor 'a' produces, the same magnetic field B_a at all points along the conductor 'b'.

Now,

A/c to Ampere's circuital law,

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$

Since, the conductor b carrying I_b will experience a sideways force due to the field B_a . The direction of this force is towards the conductor 'a'.

$$\begin{aligned} F_{ba} &= I_b L B_a \\ &= \frac{\mu_0 I_b I_a}{2\pi d} L \end{aligned}$$

Hence,

$$F_{ba} = -F_{ab}$$

OR

Establish different relations between magnetization M , magnetic intensity H , magnetic susceptibility X and relative magnetic permeability μ_r . Define the magnetic susceptibility X_s .

Sol:

We know that magnetization m of a sample to be equal to its magnetic moment per

unit volume $M = \frac{m_{net}}{V}$

Now, consider a long of n turns per unit length and carrying a current I . The magnetic field inside it is $B = \mu_0 nI$

The interior of the solenoid is filled with material with non-zero magnetization, then field inside is $B = B_0 + B_m$.

It is known as magnetic intensity H .

$$H = \frac{B}{\mu_0} - M$$

$$\therefore B = \mu_0 (H + \chi H) \quad [\because M = \chi H]$$

$$= \mu_0 H (1 + \chi)$$

$$= \mu_0 \mu_1 H$$

$$\therefore B = \mu H, \quad [\mu = \mu_0 \mu_1]$$

$$\mu_0 = 1 + \chi$$

Magnetic susceptibility (X), it is the property of material which measure of how a magnetic material responds to an external field.

Question 28:

Write Gauss's law. Determine the electric field at the points which are situated outside and inside of a uniformly charged thin spherical shell. Draw necessary diagrams of Gaussian surfaces.

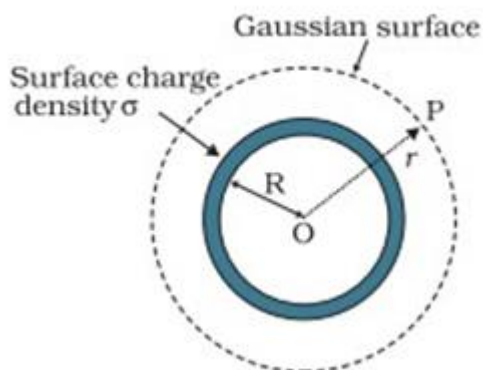
Sol:

The surface integral of the electric field intensity over any closed hypothetical surface in free space is equal to $\frac{1}{\epsilon_0}$ times the net charge enclosed within the surface. ϵ_0 is the absolute permittivity.

$$\phi_E = \oint \vec{E} \cdot d\vec{s}$$

$$= \frac{1}{\epsilon_0} \sum_{i=1}^n q_i$$

1. The point outside the shell



Let us consider a point outside the shell with radius vector r .

Now, take the Gaussian surface to be sphere of radius r with center O, passing through P. All points on this sphere are equivalent relative to the given charged configuration. The electric field at each point of the Gaussian surface, therefore, has the same magnitude E and is along the radius vector at each point. Thus, E and ΔS at every point are parallel and the flux through each element is $E \Delta S$. Summing overall ΔS , the flux through the Gaussian surface is $E \times 4\pi r^2$. The charge enclosed is $\sigma \times 4\pi R^2$.

Now,

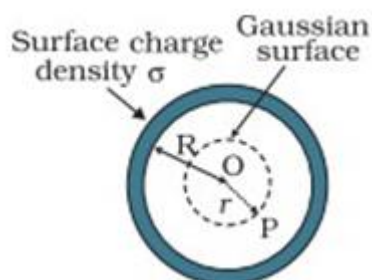
A/c Gauss's law

$$E \times 4\pi r^2 = \frac{q}{\epsilon_0} 4\pi R^2$$

$$\Rightarrow E = \frac{qR^2}{\epsilon_0 r^2}$$

$$\Rightarrow E = \frac{q}{4\pi\epsilon_0 r^2}$$

2. At point inside the shell



Here, the point is inside the shell. The Gaussian surface is again a sphere through P

centered at O. The flux through the Gaussian surface, calculated as before, is $E \times 4\pi r^2$. However; in this case, the Gaussian surface encloses no charge. Gauss's law then gives $E \times 4\pi r^2 = 0$

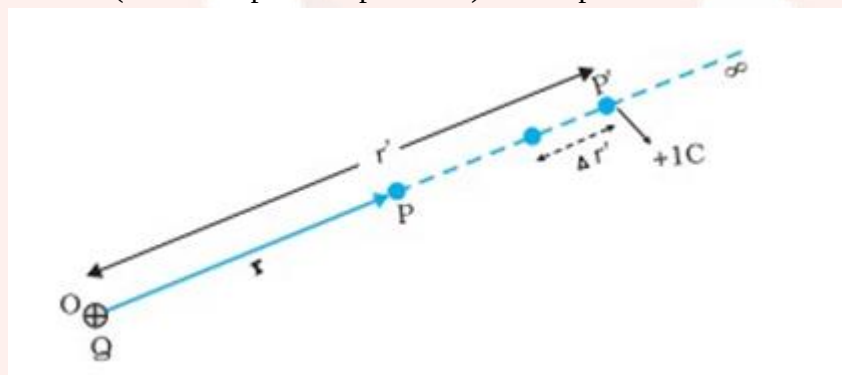
$$\therefore E = 0 (r < R)$$

OR

Write the definition of electric potential. Calculate the electric potential due to a point charge Q at a distance r from it. Draw a graph between electric potential V and distance r for a point charge Q.

Sol:

Electric potential can be defined as the amount of work done per unit positive test charge by an external agent in moving it slowly (without acceleration) from infinity (reference point of potential) to that point.



Now,

Let us consider a point charge Q at the origin. Suppose Q is positive. Such that we have to determine the potential at any point P with position vector r from the origin. The work done in bringing a unit positive test charge from infinity to the point P. For $Q > 0$, the work done against the repulsive force on the test charge is positive. Since work done is independent of the path, we choose a convenient path – along the radial direction from infinity to the point P. So, the electrostatic force on a unit positive charge is

$$\frac{Q \times 1}{4\pi\epsilon_0 r'^2}$$

Where, r' is the unit vector along OP'.

Therefore, The work done against this force from r' to $r' + \Delta r'$

$$\Delta W = -\frac{Q}{4\pi\epsilon_0 r'^2} \Delta r'$$

Here, the -ve sign shows due to $\Delta r' < Q$, ΔW is positive.

Thus, the total work done by the external force $\Delta W = -\frac{Q}{4\pi\epsilon_0 r'^2} \Delta r'$

From, $r' = \infty$ to $r' = r$

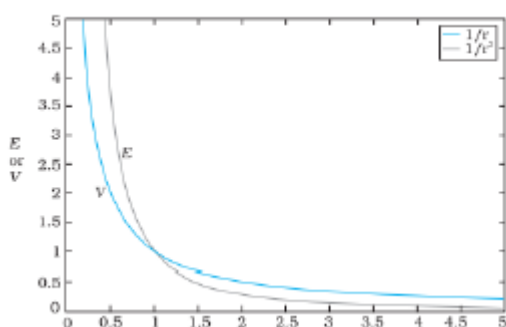
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i + \frac{d\phi_E}{dt} \right)$$

Such that

$$V(r) = \frac{Q}{4\pi r \epsilon_0}$$

Hence, the graph between electric potential V and the distance at P due to the charge Q

V(r)



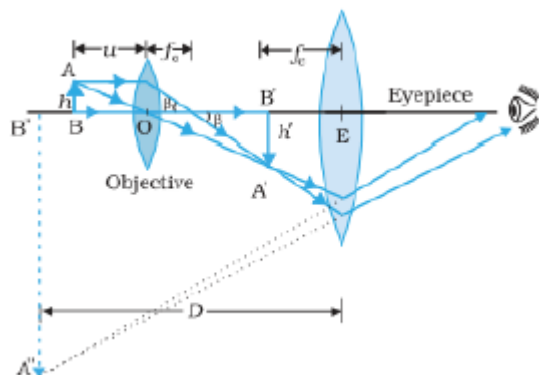
Question 29:

Describe the construction of a compound microscope. Derive an expression for its total magnification. Draw a ray diagram for the formation of image by a compound microscope.

Sol:

Microscope has a limited maximum magnification (≤ 9) for realistic focal lengths.

For much larger magnifications, one uses two lenses, one compounding the effect of the other. This is known as a compound microscope.



In this the lens nearest the object is known as objective forms a real, inverted, magnified image of the object. This serves as the object for the second lens, the eyepiece, which functions essentially like a simple microscope or magnifier, produces the final image, which is enlarged and virtual. And the final image is inverted with respect to the original object. The ray diagram shows that the (linear) magnification due to the objective

$$\therefore m_0 = \frac{h'}{h} = \frac{L}{f_0}$$

Here, h' is the size of the first image, the object size being h and f_0 the focal length of the objective. The first image is formed near the focal point of the eyepiece. The distance L , i.e., the distance between the second focal point of the objective and the first focal point of the eyepiece (focal length f_e) is called the tube length of the compound microscope.

$$m_e = \left(1 + \frac{D}{f_e}\right)$$

If the image is formed as infinity, the angular magnification

$$m_e = \frac{D}{f_e}$$

$$\therefore m = m_0 m_e$$

$$= \left(\frac{D}{f_e}\right) \left(\frac{L}{f_0}\right) \quad f$$

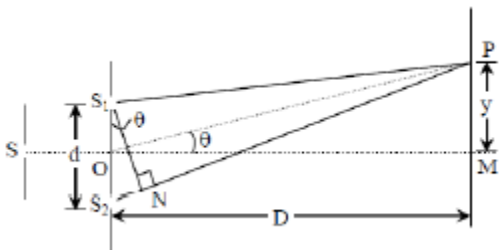
OR

What is interference? Derive the conditions to obtain constructive and destructive interferences.

If a white light source is used in place of monochromatic source of light in young's double slit experiment then what will be the effect on interference fringes.

Sol:

Interference is a phenomenon of a superposition of two or more waves and forming a resultant wave of greater amplitude or lower amplitude.



At any time t , the electric field waves at point y_1 and y_2 respectively. ϕ is the phase difference.

$$y_1 = a_1 \sin \omega t \dots (1)$$

$$y_2 = a_2 \sin (\omega t + \phi) \dots (2)$$

A/C to Young's principle of superposition.

$$y = y_1 + y_2$$

$$\Rightarrow y = a_1 \sin \omega t + a_2 \sin (\omega t + \phi)$$

$$= (a_1 + a_2 \cos \phi) \sin \omega t + (a_2 \sin \phi) \cos \omega t$$

Where,

$$A^2 = a_1^2 + a_2^2 (\cos^2 \phi + \sin^2 \phi) + 2a_1 a_2 \cos \phi$$

$$\Rightarrow A = \sqrt{a_1^2 + a_2^2 (\cos^2 \phi + \sin^2 \phi) + 2a_1 a_2 \cos \phi}$$

Now,

For constructive interference

Maximum intensity at any point $\cos \phi = +1$

Then the phase difference

$$\phi = 0, 2\pi, 4\pi, \dots, 2n\pi (n = 0, 1, 2, \dots)$$

And,

Path difference

$$\Delta x = 0, \lambda, 2\lambda, \dots, n\lambda$$

Time difference

$$\Delta t = 0, T, 2T, 3T, \dots, nT$$

Destructive interference

Maximum intensity at any point $\cos \phi = -1$

Phase difference

$$\phi = (2n-1)\pi (n = 0, 1, 2, 3, \dots)$$

And,

Path difference

$$\Delta x = (2n-1)\frac{\lambda}{2}$$

Time difference

$$\Delta t = (2n-1)\frac{T}{2}$$

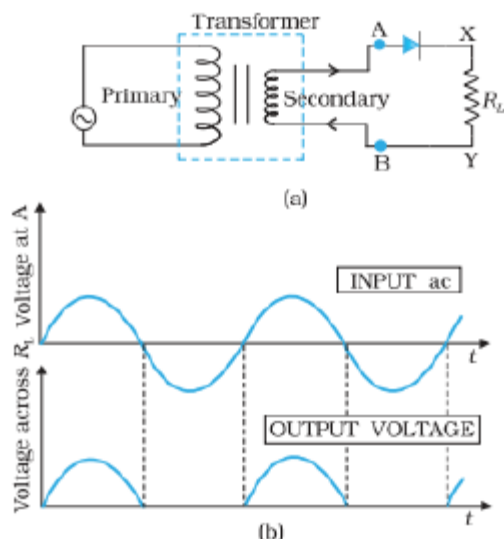
The interference pattern due to different colour component of white light overlap. The central bright fringes for different colours are at the same position. Such that central fringes are white.

Question 30:

What is rectification? Draw the circuit diagram of half wave rectifier and explain its working. Show the input ac voltage and output voltage waveforms from the rectifier circuit.

Sol:

Rectification is the process of converting bi directional current flow to unidirectional current flow.



If an alternating voltage is applied across a diode in series with a load, a pulsating voltage will appear across the load only during the half cycles of the ac input during which the diode is forward biased. Such rectifier circuit is called a half-wave rectifier. The secondary of a transformer supplies the desired ac voltage across terminals A and B. When the voltage at A is positive, the diode is forward biased and it conducts. When A is negative, the diode is reverse-biased and it does not conduct. The reverse saturation current of a diode is negligible and can be considered equal to zero for practical purposes.

OR

What is extrinsic semiconductor? How many types of these are? Write their names. Explain the processes which are occurred during the formation of a P-N junction.

Determine the electric field produced at a P-N junction when width of depletion layer is 1 micrometer and barrier potential is 0.7 volt

Sol:

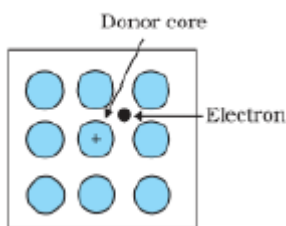
The electric field product at P-N junction $\frac{V}{d}$

$$d = 10^{-6} \text{ m and } V = 0.7 \text{ in } \frac{V}{d}$$

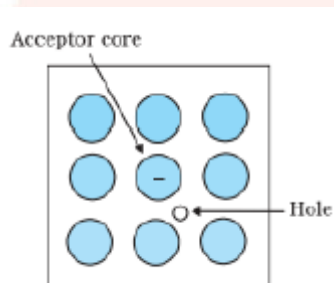
$$\frac{V}{d} = \frac{10^{-6}}{0.7} = 1.43 \times 10^{-6} \text{ V / m}$$

When a small amount, say, a few parts per million (ppm), of a suitable impurity is added to the pure semiconductor, the conductivity of the semiconductor is increased manifold. Such materials are known as extrinsic semiconductors or impurity semiconductor.

n-Type: An extrinsic semiconductor doped with pentavalent impurity, electrons become the majority carriers and holes the minority carriers. These semiconductors are, therefore, known as n-type semiconductors.



p-Type: The Extrinsic semiconductors doped with trivalent impurity are called p-type semiconductors. For p-type semiconductors, the recombination process will reduce the number (n_i) intrinsically generated electrons.



(i) Diffusion:

In n-type semiconductor, the concentration of electrons is much greater as compared to concentration of holes; while in p-type semiconductor, the concentration of holes is much greater than the concentration of electron. When a p-n junction is formed, then due to concentration gradient, the holes diffuse from p side to n side ($p \rightarrow n$) and electrons diffuse from n side to p-side ($n \rightarrow p$).

(ii) Drift

The drift of charge carriers occurs due to electric field. Due to built in potential barrier an electric field directed from n-region to p-region is developed across the junction. This field causes motion of electrons on p-side of the junction to n-side and motion of holes on n-side of junction to p-side. Thus a drift current starts.